

5.6 解: 设刚体转动角速度 $\vec{\omega} = \omega_i \vec{i} + \omega_j \vec{j} + \omega_k \vec{k}$

$$\vec{v}_i = \vec{\omega} \times \vec{op}_i = -\omega_k \vec{i} - \omega_j \vec{j} + (\omega_i + \omega_j) \vec{k}$$

$$P_2 \text{ 处 } \vec{v}_2 = \vec{\omega} \times \vec{op}_2 = \omega_j \vec{i} - \omega_k \vec{j}$$

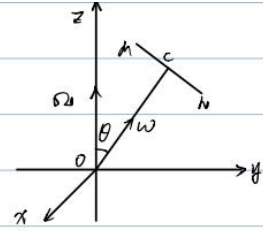
$$\text{则有 } \begin{cases} v_1 = \sqrt{\omega_k^2 + \omega_j^2 + (\omega_i + \omega_j)^2} = 2 \\ \vec{v}_1 \cdot \vec{i} = -\omega_k = v_1 \cos 60^\circ \\ v_2 \cdot \vec{i} = \omega_j = v_2 = \sqrt{\omega_j^2 + \omega_k^2} \end{cases}$$

$$\text{解得: } \vec{\omega} = 2\vec{j} - \vec{k}$$

$$\vec{v}_2 = 2\vec{i}$$

$$\text{转动瞬轴: } \begin{cases} y = 2t \\ z = -t \end{cases} \quad x, t \in \mathbb{R}$$

5.16 解:



建立固结在O点的坐标系 $Oxyz$.

$$AB \text{ 转动角速度 } \vec{\omega}_1 = k \text{ rad/s}$$

$$MN \text{ 绕 } OC \text{ 转动角速度 } \vec{\omega} = 10 \sin \theta \vec{j} + 10 \cos \theta \vec{k} \text{ rad/s}$$

$$OC \text{ 绕 } AB \text{ 转动角速度 } \vec{\omega}_0 = -\frac{3}{2} \vec{i} \text{ rad/s}$$

$$\text{圆盘总角速度 } \vec{\omega}_2 = \vec{\omega} + \vec{\omega}_1 + \vec{\omega}_0$$

$$= -\frac{3}{2} \vec{i} + 10 \sin \frac{3\pi}{4} \vec{j} + (1 + 10 \cos \frac{3\pi}{4}) \vec{k} \text{ rad/s}$$

$$\text{角加速度 } \vec{\epsilon} = 5\sqrt{2} \cos \frac{3\pi}{4} \vec{j} - 5\sqrt{2} \sin \frac{3\pi}{4} \vec{k}$$

$$t = 1 \text{ s 时, } \vec{\omega}_1 = -\frac{3}{2} \vec{i} + 10 \vec{j} + \vec{k} \quad \vec{\epsilon} = -5\sqrt{2} \vec{k}$$

$$\vec{om} = 10 \vec{j} + 5 \vec{k} \text{ cm} \quad \vec{on} = 10 \vec{j} - 5 \vec{k} \text{ cm}$$

$$M \text{ 点速度 } \vec{v}_m = \vec{\omega}_1 \times \vec{om} = 60 \vec{i} + \frac{5}{2} \vec{j} - 5\sqrt{2} \vec{k} \text{ cm/s}$$

$$\text{加速度 } \vec{a}_m = \vec{\epsilon} \times \vec{om} + \vec{\omega}_1 \times \vec{v}_m$$

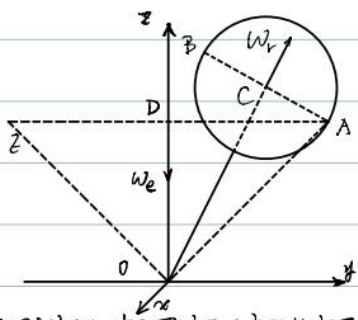
$$= -\frac{5\sqrt{2}}{2} \vec{i} + (40 - \frac{5\sqrt{2}}{2}) \vec{j} - (\frac{5\sqrt{2}}{2} + 600) \vec{k} \text{ cm/s}^2$$

$$N \text{ 点速度 } \vec{v}_n = \vec{\omega}_1 \times \vec{on} = -60 \vec{i} - \frac{5\sqrt{2}}{2} \vec{j} - 5\sqrt{2} \vec{k}$$

$$\text{加速度 } \vec{a}_n = \vec{\epsilon} \times \vec{on} + \vec{\omega}_1 \times \vec{v}_n$$

$$= \frac{5\sqrt{2}}{2} \vec{i} - (\frac{5\sqrt{2}}{2} + 60) \vec{j} + (\frac{5\sqrt{2}}{2} + 600) \vec{k} \text{ cm/s}^2$$

5.11 解



刚体作规则运动，建立固结于O点的坐标系

$$\left. \begin{aligned} OD \tan \theta + \frac{AC}{\cos \theta} &= AD = OD \\ \frac{OD}{\cos \theta} + AC \tan \theta &= OC = 3AC \end{aligned} \right\} \Rightarrow \begin{cases} \tan \theta = \frac{1}{2} \\ OD = 5\sqrt{5} \end{cases}$$

刚体公转角速度满足

$$\vec{\omega}_e \times (\vec{OC} \cdot \vec{j}) = \vec{v} = 15 \vec{i}$$

$$\vec{\omega}_e = -\sqrt{5} \vec{k} \text{ (rad/s)}$$

$$A \text{ 点速度 } \vec{v}_A = \vec{\omega}_e \times \vec{OA} + \vec{\omega}_r \times \vec{CA} = 0$$

$$\vec{\omega}_r = \sqrt{5} \vec{j} + 2\sqrt{5} \vec{k} \text{ (rad/s)}$$

$$\vec{\omega} = \vec{\omega}_e + \vec{\omega}_r = \sqrt{5} \vec{j} + \sqrt{5} \vec{k} \text{ (rad/s)}$$

故刚体瞬时角加速度为

$$\vec{\epsilon} = \vec{\omega}_e \times \vec{\omega}_r = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\sqrt{5} \\ 0 & \sqrt{5} & \sqrt{5} \end{vmatrix} = 5 \vec{i}$$

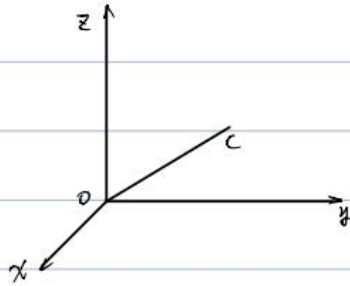
AO为转动瞬轴，故A点的加速度为

$$\vec{a}_A = \vec{\epsilon} \times \vec{OA} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 5 & 0 & 0 \\ 0 & 5\sqrt{5} & 5\sqrt{5} \end{vmatrix} = -25\sqrt{5} \vec{j} + 25\sqrt{5} \vec{k} \text{ (cm/s}^2\text{)}$$

$$OB = OA = 5\sqrt{5} \quad \vec{OB} = \sqrt{5} \vec{j} + 7\sqrt{5} \vec{k}$$

$$B \text{ 点加速度 } \vec{a}_B = \vec{\epsilon} \times \vec{OB} + \vec{\omega} \times (\vec{\omega} \times \vec{OB}) = -5\sqrt{5} \vec{j} - 25\sqrt{5} \vec{k} \text{ (cm/s}^2\text{)}$$

5.17 解:



以O为基点建立坐标系 $Oxyz$, xOz 平面与 ABD 平面重合

$$\text{则 } \vec{\omega}_1 = \omega_1 \vec{k} \quad \vec{\omega}_2 = \frac{\sqrt{3}}{2} \omega_2 \vec{j} + \frac{1}{2} \omega_2 \vec{k}$$

$$\text{MN的角速度 } \vec{\omega} = \vec{\omega}_1 + \vec{\omega}_2 = \frac{\sqrt{3}}{2} \omega_2 \vec{j} + (\omega_1 + \frac{1}{2} \omega_2) \vec{k}$$

$$\text{角加速度 } \vec{\varepsilon} = \vec{\omega}_1 \times \vec{\omega}_2 = -\frac{3}{2} \omega_1 \omega_2 \vec{i}$$

$$\vec{v}_O = -\omega_1 d \vec{j} \quad \vec{a}_O = \omega_1^2 d \vec{i}$$

$$\vec{om} = (\sqrt{3} - \frac{1}{2}) r \vec{j} + (1 + \frac{\sqrt{3}}{2}) r \vec{k} \quad \vec{on} = (\sqrt{3} + \frac{1}{2}) r \vec{j} + (1 - \frac{\sqrt{3}}{2}) r \vec{k}$$

$$\text{故 } \vec{v}_m = \vec{v}_O + \vec{\omega} \times \vec{om} = [\omega - (\sqrt{3} - \frac{1}{2}) \omega_1] r \vec{i} + \omega_1 d \vec{j}$$

$$\vec{a}_m = \vec{a}_O + \vec{\varepsilon} \times \vec{om} + \vec{\omega} \times \vec{v}_m$$

$$= (2\omega_1^2 d + \frac{1}{2} \omega_1 \omega_2 d) \vec{i} + [\frac{1}{2} \omega^2 - (\sqrt{3} - \frac{1}{2}) \omega_1^2 + 2\omega \omega_1] \vec{j} - \frac{\sqrt{3}}{2} \omega^2 r \vec{k}$$

$$\vec{v}_n = \vec{v}_O + \vec{\omega} \times \vec{on} = [-\omega - (\sqrt{3} + \frac{1}{2}) \omega_1] r \vec{i} - \omega_1 d \vec{j}$$

$$\vec{a}_n = \vec{a}_O + \vec{\varepsilon} \times \vec{on} + \vec{\omega} \times \vec{v}_n$$

$$= (2\omega_1^2 d + \frac{1}{2} \omega_1 \omega_2 d) \vec{i} - [\frac{1}{2} \omega^2 + (\sqrt{3} + \frac{1}{2}) \omega_1^2 + 2\omega \omega_1] \vec{j} + \frac{\sqrt{3}}{2} \omega^2 r \vec{k}$$

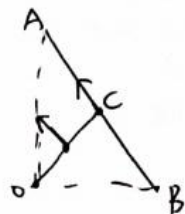
6-3. AB 杆是对称的, 则 AB 杆的质心在 C 点, 且 $v_C = \omega l$

$$\therefore p_1 = m v_C = 2(m_1 + m_2) \cdot \omega l$$

OC 杆也是均质的, 其质心在 OC 的中点

$$\therefore p_2 = m_1 v_{OC \text{ 质心}} = m_1 \cdot \frac{1}{2} \omega l$$

$$p_1, p_2 \text{ 同向. } \therefore p = p_1 + p_2 = (2.5m_1 + m_2) \omega l$$



6.12 解: 取 P_1, P_2, P_3 和四棱柱 ABCD 为分离体

质点系所受重力和支持力均沿竖直方向

水平方向不受外力, 故质心位置不变

P_1 下降高度 $h=1\text{m}$ 时, 设四棱柱位移为 x , 则

$$m_0 x + m_1 x + m_2 (x-h) + m_3 (x-h \cos 60^\circ) = 0$$

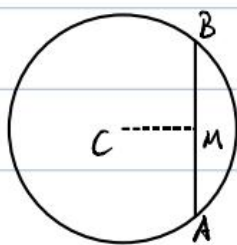
$$\text{解得 } x = \frac{4}{29} \text{ m} \approx 0.138 \text{ m}$$

四棱柱 ABCD 向左移动 0.138 m

6.18 解: 选点 M 和圆盘为分离体

初始时质点系角动量

$$H_z = \frac{Pr^2\omega}{2g} + \frac{Qr\omega}{g}$$



质点系运动过程中不受外力, 角动量守恒

当 M 离 C 最近时在 AB 中点, 设圆盘角速度 ω'

$$M \text{ 速度 } v_M = \omega' a + u$$

$$\text{故有 } H_z = \frac{Pr^2\omega'}{2g} + \frac{Qa}{g} (\omega' a + u)$$

$$\text{解得: } \omega' = \frac{(P+2Q)r^2\omega - 2Qau}{Pr^2 + 2Qa^2}$$

6.24 解: 设 A、B 受斜面的摩擦为 f_A, f_B 则

$$\text{对 A: } ma = mg \sin \alpha - f_A - N$$

$$7mr^2\varepsilon = r(f_A - f_N)$$

$$\text{对 B: } ma = mg \sin \alpha + N - f_B$$

$$mr^2\varepsilon = r(f_B - f_N)$$

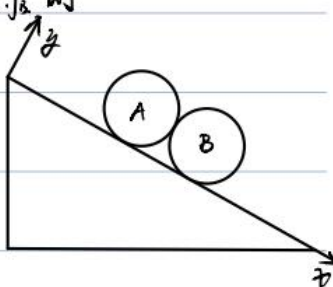
$$f_B = f_A + 2N_{AB}$$

$$f_A + N_{AB} = f_B - N_{AB}$$

$$\text{另有: } \varepsilon r = a$$

$$\text{解得: } a = \frac{2g \sin \alpha}{3 + f + 7 - f}$$

$$N = \frac{(1 - \gamma)mg \sin \alpha}{3 + f + 7 - f}$$



6.44 解: 以圆柱体为研究对象, 其转动惯量 $I = \frac{1}{2}mR^2$

受外力: mg, T

$$\text{故有: } m\ddot{h} = mg - T$$

$$I\varepsilon = RT$$

$$\varepsilon R = \ddot{h}$$

$$\text{解得: } \ddot{h} = \frac{2}{3}g \quad h = \frac{1}{3}gt^2 \quad v = \frac{2}{3}gt$$

$$\text{故下降 } h \text{ 时 } t = \sqrt{\frac{3h}{g}}$$

$$\text{速度 } v = 2\sqrt{\frac{gh}{3}} \quad \text{绳拉力 } T = \frac{1}{3}mg$$

6. (2) 解: (1) 对球, 转动惯量 $I = \frac{2}{5} mR^2$

$$m\ddot{x} = mg \sin \alpha - f_1$$

$$I\epsilon = Rf_1$$

$$R\epsilon = \ddot{x}$$

解得: $\ddot{x} = \frac{5}{7} g \sin \alpha$

$$x = \frac{5}{14} g t^2 \sin \alpha \quad \dot{x} = \frac{5}{7} g t \sin \alpha$$

当 $x \sin \alpha = h$ 时 $t = \sqrt{\frac{14h}{5g \sin \alpha}}$

此时速度 $v = \dot{x} = \sqrt{\frac{10gh}{7}}$

摩擦力 $f_1 = \frac{2}{7} mg \sin \alpha$

(2) 对圆柱 转动惯量 $I = \frac{1}{2} mR^2$

$$m\ddot{x} = mg \sin \alpha - f_2$$

$$I\epsilon = Rf_2$$

$$R\epsilon = \ddot{x}$$

解得: $\ddot{x} = \frac{2}{3} g \sin \alpha$

$$x = \frac{1}{3} g t^2 \sin \alpha \quad \dot{x} = \frac{2}{3} g t \sin \alpha$$

当 $x \sin \alpha = h$ 时 $t = \sqrt{\frac{3h}{g \sin \alpha}}$

此时速度 $v = \dot{x} = \sqrt{\frac{4gh}{3}}$

摩擦力 $f_2 = \frac{1}{3} mg \sin \alpha$

(3) 对圆环, 转动惯量 $I = mr^2$

同理得: $\ddot{x} = \frac{1}{2} g \sin \alpha$

$$x = \frac{1}{4} g t^2 \sin \alpha \quad \dot{x} = \frac{1}{2} g t \sin \alpha$$

当 $x \sin \alpha = h$ 时 $t = \sqrt{\frac{4h}{g \sin \alpha}}$

此时速度 $v = \dot{x} = \sqrt{gh}$

摩擦力 $f_3 = \frac{1}{2} mg \sin \alpha$

6.60解: 设角速度为 ω , 则 AB 杆速度 $v_A = \omega a$

$$OC \text{ 转动惯量 } I_0 = \frac{P_1 l^2}{3g}$$

整个运动过程只有 P_1, P_2 重力做功, 由动能定理

$$\frac{P_1 l}{2} \sin \varphi_0 + P_2 a \tan \varphi_0 = \frac{1}{2} I_0 \omega^2 + \frac{1}{2} \frac{P_2}{g} v_A^2$$

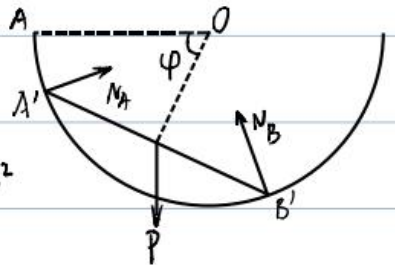
$$\text{解得: } \omega = \sqrt{\frac{3gP_1 l \sin \varphi_0 + 6gP_2 a \tan \varphi_0}{P_1 l^2 + 3P_2 a^2}}$$

6.69解: AB 对 O 点

转动惯量

$$I_0 = \frac{4}{12} m a^2 + m a^2$$

$$= \frac{4}{3} \frac{P}{g} a^2$$



运动过程重力做功, 由动能定理

$$P(a \sin \varphi - a \sin 45^\circ) = \frac{1}{2} I_0 \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{3g(2 \sin \varphi - \sqrt{2})}{4a}}$$

对 AB 受力分析 $a_n = \omega^2 a$ $a_t = \varepsilon a$

$$\frac{P}{g} a_n = \frac{\sqrt{2}}{2} (N_A + N_B) - P \sin \varphi$$

$$\frac{P}{g} a_t = \frac{\sqrt{2}}{2} (N_A - N_B) + P \cos \varphi$$

$$I_0 \varepsilon = a P \cos \varphi$$

$$\text{解得: } N_A = \left(\frac{5\sqrt{2}}{4} \sin \varphi - \frac{\sqrt{2}}{8} \cos \varphi - \frac{3}{4} \right) P$$

$$N_B = \left(\frac{5\sqrt{2}}{4} \sin \varphi + \frac{\sqrt{2}}{8} \cos \varphi - \frac{3}{4} \right) P$$

6.76解: 力偶做功 $W_1 = \int_0^{\frac{\pi}{2}} M d\varphi = \frac{11M_1\pi}{32}$

由动能定理:

$$W_1 - m_2 g \frac{\sqrt{2}}{2} b - m_1 g l - \frac{1}{2} k l^2 = \frac{1}{2} I_0 \omega^2 + \frac{1}{2} m_1 v_B^2$$

$$v_B = \omega \cdot \sqrt{2} l \cdot \sqrt{2}$$

解得: $\omega = \sqrt{\frac{\frac{11}{16} M_1 \pi - 2m_1 g l - \sqrt{2} m_2 g b - k l^2}{I_0 + 4m_1 l^2}}$

6.80解: I的转动惯量 $I_A = \frac{1}{2} m_1 r_1^2$

II的转动惯量 $I_B = \frac{1}{2} m_2 r_2^2$

III的转动惯量 $I_C = \frac{1}{2} m_3 r_3^2$

设轮II侧绳拉力为 T_2 , 轮III侧绳拉力为 T_3

对轮I $I_A \varepsilon_1 = r_1 (T_2 - T_3)$

对轮II $m_2 a_2 = m_2 g - T_2$

$$I_B \varepsilon_2 = r_2 T_2$$

$$\varepsilon_2 r_2 = a_2 - \varepsilon_1 r_1$$

对轮III $m_3 a_3 = m_3 g - T_3$

$$I_C \varepsilon_3 = r_3 T_3$$

$$\varepsilon_3 r_3 = \varepsilon_1 r_1 + a_3$$

解得: 轮I角加速度: $\varepsilon_1 = \frac{2(m_2 - m_3)g}{(3m_1 + 2m_2 + 2m_3)r_1}$

方向垂直纸面向外

轮II质心加速度 $a_2 = \frac{6m_1 + 6m_2 + 2m_3}{3(3m_1 + 2m_2 + 2m_3)} g$

方向竖直向下

轮III质心加速度 $a_3 = \frac{6m_1 + 2m_2 + 6m_3}{3(3m_1 + 2m_2 + 2m_3)} g$

方向竖直向下